# Quasi-Poisson models

# Last time: modeling article publication

We are interested in analyzing the number of articles published by biochemistry PhD students. The data contains the following variables:

- `art`: articles published in last three years of Ph.D.
- `kid5`: number of children under age six
- `phd`: prestige of Ph.D. program
- `ment`: articles published by their mentor in last three years

# Activity

Work on the class activity (course website).

# Recap: confidence interval

Poisson output:

| | Estimate | Std. Error | z value | Pr(>\|z\| |
|---|---|---|---|---|
| (Intercept) | 0.276493292 | 0.087294456 | 3.1673637 | 1.538278e-0 |
| phd | 0.008071036 | 0.026357684 | 0.3062119 | 7.594433e-0 |
| ment | 0.026537317 | 0.002000733 | 13.2637987 | 3.753884e-4 |
| kid5 | -0.099696463 | 0.035248105 | -2.8284205 | 4.677832e-0 |

Want: 95% confidence interval for the change in the expected number of articles published associated with an increase of one in the number of children under age 6, holding PhD prestige and mentor publications fixed.

for $\beta_3$ : $-0.0997 \pm 1.96(0.03525) = (-0.169, -0.031)$

for $e^{\beta_3}$ : $(e^{-0.169}, e^{-0.031}) = (0.845, 0.969)$

We are 95% confident that an additional child under age 6 is associated w/ a decrease in average # of articles by between 3.1% and 15.5%, holding PhD prestige and mentor publications fixed

# Activity: comparing standard errors

- Poisson estimate: $\widehat{SE}(\hat{\beta}_3)_{\text{poisson}} = 0.03525$
- Bootstrap estimate: $\widehat{SE}(\hat{\beta}_3)_{\text{boot}} = 0.0490$

> What will be the effect if we use the Poisson SE, when the Poisson distribution is not a good choice?

$\hat{SE}_{poisson}$ is too small

↳ CIs are too narrow!

(a "95%" confidence won't actually have desired 95% coverage, b/c it is too narrow)

# Overdispersion

**Overdispersion** occurs when the response $Y$ has higher variance than we would expect if $Y$ followed a Poisson distribution.

Let

dispersion parameter

$$\phi = \frac{\text{Variance}}{\text{Mean}}$$

(Specific to a Poisson)

> What should $\phi$ be if there is no overdispersion?

If Poisson is correct, then mean = variance

$\Rightarrow \phi = 1$

overdispersion means $\phi > 1$

# Estimating overdispersion

The *Pearson residual* for observation $i$ is defined as

$$e_{(P)i} = \frac{Y_i - \widehat{\lambda}_i}{\sqrt{\widehat{\lambda}_i}}$$

← observed − predicted

← std dev. of $Y$ (if Poisson is correct)

$$\widehat{\phi} = \frac{\sum_{i=1}^{n} e_{(P)i}^2}{n-p}$$

← numerator would be $\approx n-p$ if Poisson model is correct

↑ residual df

- $p =$ number of parameters in model

If Poisson is a good choice, $\hat{\phi} \approx 1$

Otherwise we might see $\hat{\phi} > 1$

# Example: Estimating overdispersion

```
# fit the model
m1 <- glm(art ~ phd + ment + kid5,
          data = articles, family = poisson)

# get Pearson residuals
pearson_resids <- resid(m1, "pearson")

# estimate dispersion parameter
phihat <- sum(pearson_resids^2)/(915 - 4)
phihat
```

```
## [1] 1.888732
```

Variance ≈ 1.89 times mean

# Handling overdispersion

Overdispersion is a problem because our standard errors (for confidence intervals and hypothesis tests) are too low.

> If we think there is overdispersion, what should we do?

- option 1: Bootstrap!
- option 2: use $\hat{\phi}$ to adjust SEs in fitted model
- option 3: use a different distribution (e.g., negative binomial)

# Adjusting the standard error

- In our data, $\widehat{\phi} = 1.89$
- This means our variance is 1.89 times bigger than it should be
- So our standard error is $\sqrt{1.89} = 1.375$ times bigger than it should be

# Adjusting the standard error in R

```
m2 <- glm(art ~ phd + ment + kid5, data = articles,
          family = quasipoisson)
```

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.276493   0.119839   2.307   0.0213 *
phd          0.008071   0.036184   0.223   0.8235
ment         0.026537   0.002747   9.662   <2e-16 ***
kid5        -0.099696   0.048389  -2.060   0.0397 *

(Dispersion parameter for quasipoisson family taken to be 1.88461)
```

- Allowing $\phi$ to be different from 1 means we are using a *quasi-likelihood* (in this case, a *quasi-Poisson*)

# Adjusting the standard error

```
glm(art ~ phd + ment + kid5, family = poisson, data = articles)
```

```
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.276493   0.087294   3.167  0.00154 **
phd          0.008071   0.026358   0.306  0.75944
ment         0.026537   0.002001  13.264  < 2e-16 ***
kid5        -0.099696   0.035248  -2.828  0.00468 **
(Dispersion parameter for poisson family
   taken to be 1)
```

$\hat{\beta}$ are the same!

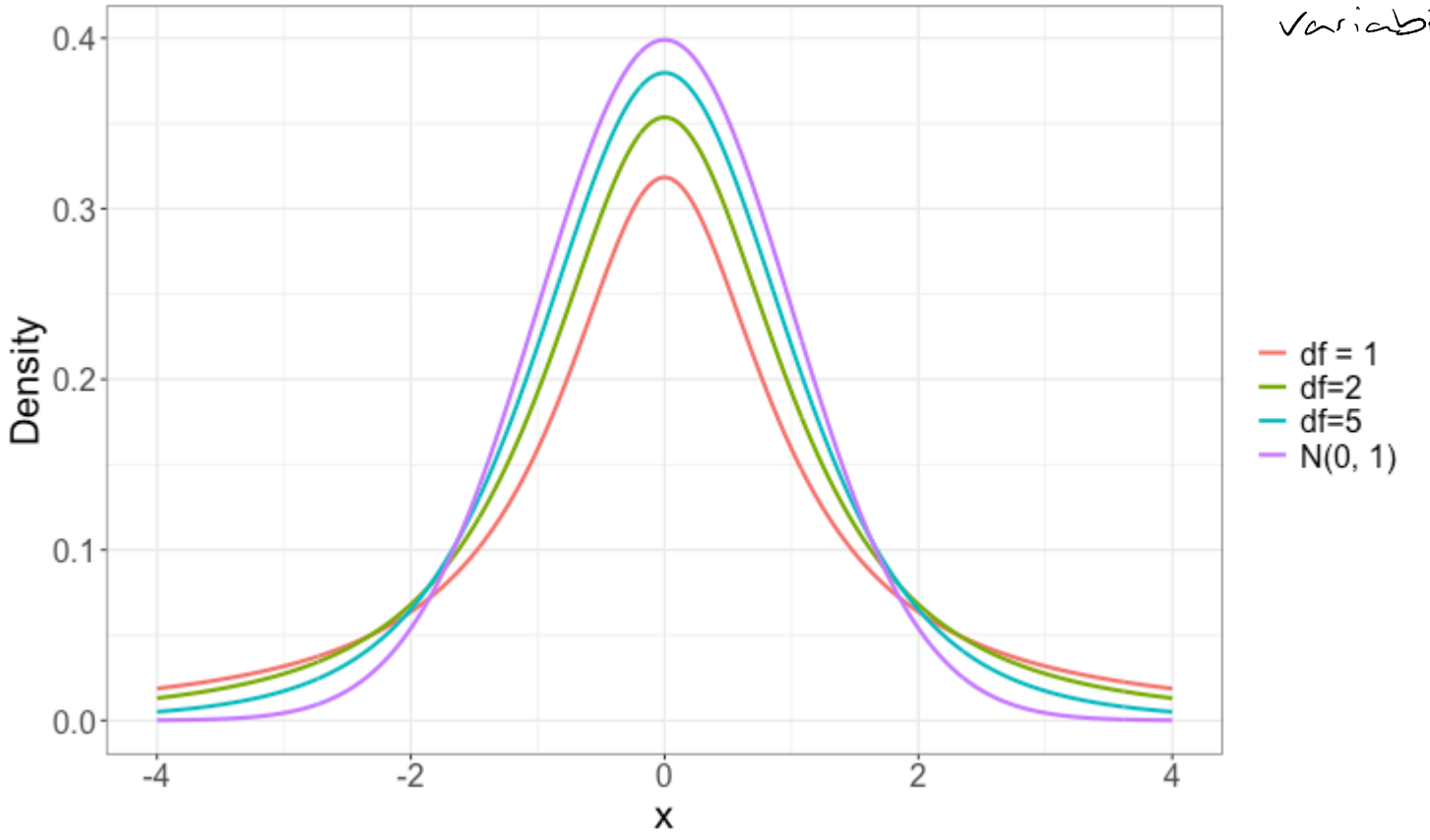
```
glm(art ~ phd + ment + kid5, family = quasipoisson, data = articles)
```

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.276493   0.119839   2.307   0.0213 *
phd          0.008071   0.036184   0.223   0.8235
ment         0.026537   0.002747   9.662   <2e-16 ***
kid5        -0.099696   0.048389  -2.060   0.0397 *
(Dispersion parameter for quasipoisson family
   taken to be 1.88461)
```

$\hat{\phi}$ $\quad 0.0484 = \sqrt{\hat{\phi}}\ (0.0352)$

# t-distribution

Intuition: estimating $\sigma$ => more variability in estimated statistics => t distribution (which has greater variability)



# Calculating a confidence interval

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.276493   0.119839   2.307   0.0213 *
phd          0.008071   0.036184   0.223   0.8235
ment         0.026537   0.002747   9.662   <2e-16 ***
kid5        -0.099696   0.048389  -2.060   0.0397 *
```

New confidence interval for $\beta_3$:

$$-0.0997 \pm t^*_{n-p} \cdot 0.0484$$

```
qt(0.025, df=909, lower.tail=F)
```

```
## [1] 1.962577
```

$t^*_{909}$

$$-0.0997 \pm 1.963 \cdot 0.0484 = (-0.195,\ 0.005)$$